COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 6

10 Logic and Proof (lp15)

clause methods, DPLL, resolution theorem proving

(*a*) List three significant differences between the DPLL method and resolution. [3 marks]

Answer: Possible answers include DPLL is for propositional logic while resolution is for first-order logic; DPLL uses case-splitting while resolution uses saturation; DPLL is a decision procedure while resolution is not; resolution uses unification but DPLL does not; DPLL scales to enormous problems; resolution generates new clauses while DPLL only instantiates variables in clauses.

first-order logic, resolution

(*b*) To what extent is pure literal elimination applicable to (i) resolution theorem proving and (ii) Prolog? [3 marks]

Answer: Part of DPLL, pure literal elimination is the removal of all clauses containing a pure literal, i.e. one that appears everywhere with the same sign (positive or negative but not both, ignoring the arguments). The same concept applies to resolution, where it would remove clauses (presumably from an included axiom file) irrelevant to the theorem being proved.

In Prolog, a pure literal is either code that is never called or a predicate that is called but never defined. The former is dead code that can be removed and the latter could be flagged by the Prolog compiler.

first-order logic, resolution

(*c*) For each of the following sets of clauses, either exhibit a model or show that none exists using resolution. Note: a and b are constants, while x and y are variables.

(*i*)

$$\{P(a)\} \qquad \{\neg P(x), M(x)\} \qquad \{\neg P(x), L(x)\} \qquad \{\neg Q(x), \neg P(x)\}$$
$$\{\neg L(x), \neg M(x), \neg R(x)\} \qquad \{\neg P(x), Q(x), P(b)\} \qquad \{\neg P(x), Q(x), R(b)\}$$

[7 marks]

Answer: Resolving the unit clause $\{P(a)\}$ successively with the other clauses, we derive $\{M(a)\}$, $\{L(a)\}$, $\{\neg Q(a)\}$, $\{Q(a), P(b)\}$, $\{Q(a), R(b)\}$. From the last three of these we derive $\{P(b)\}$ and $\{R(b)\}$. From $\{M(a)\}$ and $\{L(a)\}$ we derive $\{\neg R(a)\}$.

Repeating the above using $\{P(b)\}$ we derive $\{M(b)\}$, $\{L(b)\}$ and hence $\{\neg R(b)\}$, contradicting $\{R(b)\}$ and yielding the empty clause.

(*ii*)

$$\{\neg P(x), Q(x,x)\} \qquad \{\neg Q(x,y), \neg Q(y,x), R(x,y)\}$$
$$\{\neg R(x,y), \neg R(a,x)\} \qquad \{P(a), P(b)\}$$

[7 marks]

Answer: One might try resolution on this problem, deriving $\{\neg R(a,a)\}$ by factoring the third clause. Resolving this with the second clause yields $\{\neg Q(a,a)\}$ and thence $\{\neg P(a)\}$.

But now things look stuck, because $P(b)$ could be true. In fact, if P, Q, R are defined to be true only in the cases $P(b)$, $Q(b,b)$ and $R(b,b)$ then it is easy to see that all four clauses are satisfied. The model might use the natural numbers for the universe (we must have $a \neq b$, so we need at least two elements) and we could put simply $a = 1$ and $b = 0$, etc. The resolution steps we performed show that $P(a)$, $Q(a,a)$ and $R(a,a)$ must be false in any model.